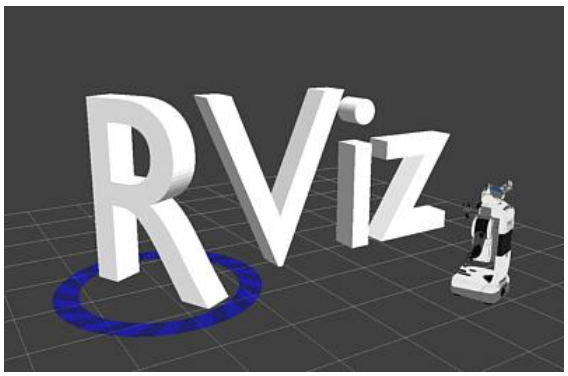
ROS2 4-DOF Industrial Robotic Arm





By,

Muhammed Sinan P

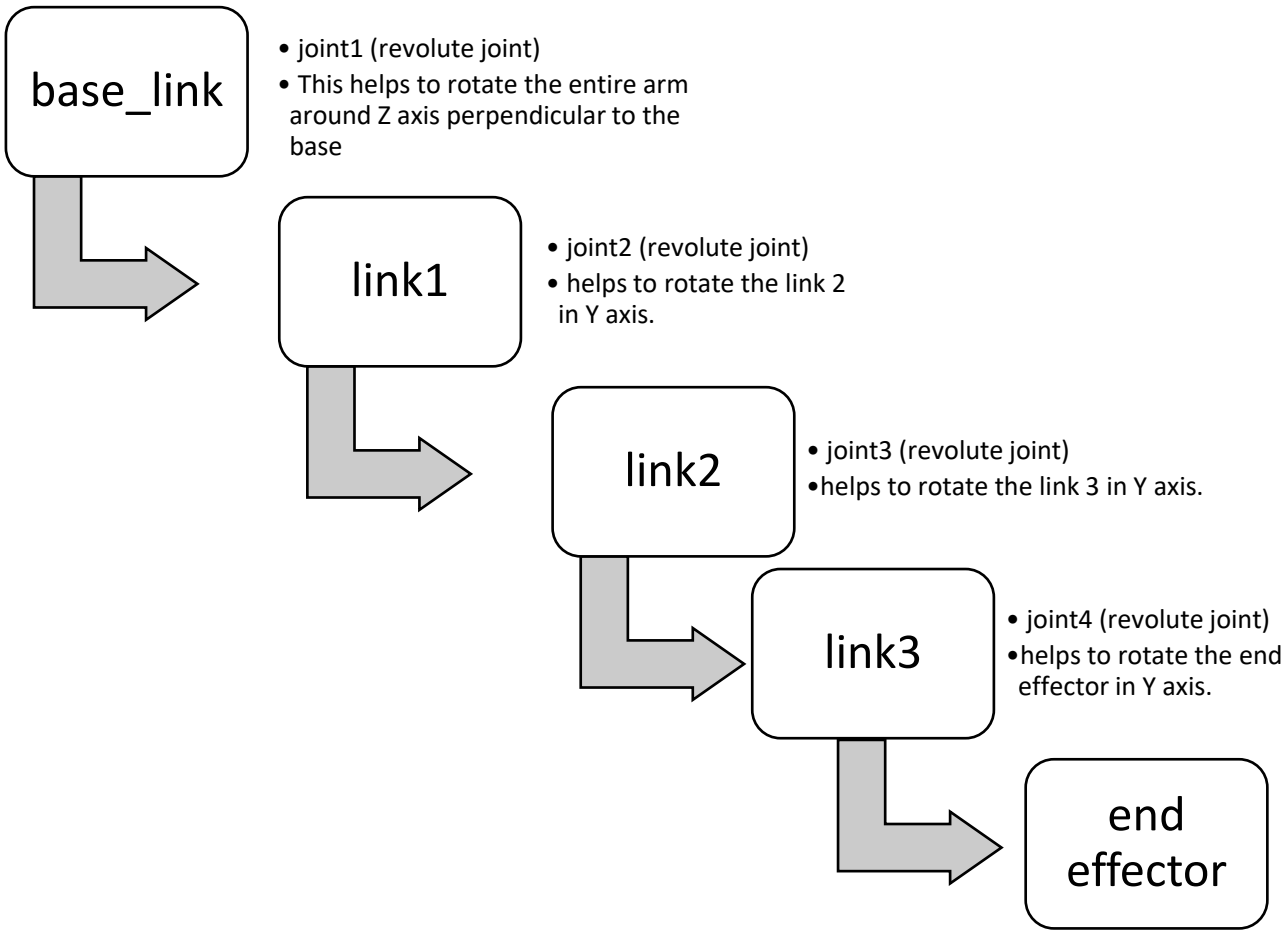
Team Alexandria

Github: https://github.com/sinanp1x/ros2-4dof-industrial-arm.git

Question 1 – Robot Structure Design

The design of this 4-dof arm is really inspired from the design structure of “Universal Robots UR5”, but designed using basic rectangle boxes to mimic the design.

The link and joint tree are like:



Question 2 – URDF Link Definition

The URDF file was created from the well described XACRO file, which helps better to reduce repeated typing and mistakes.

XACRO FILE:

```
<robot name="four_dof_arm" xmlns:xacro="http://www.ros.org/wiki/xacro">

    <xacro:property name="baseparam" value="1.0 1.0 0.3"/>
    <xacro:property name="linkparam" value="0.2 0.2 1.0"/>
    <xacro:property name="efparam" value="0.2 0.2 0.5"/>

    <xacro:macro name="build_link" params="name size xyz">
```

```
        <link name="${name}">
            <visual>
                <origin xyz="${xyz}" rpy="0 0 0"/>
                <geometry>
                    <box size="${size}"/>
                </geometry>
            </visual>
            <collision>
                <origin xyz="${xyz}" rpy="0 0 0"/>
                <geometry>
                    <box size="${size}"/>
                </geometry>
            </collision>
        </link>
    </xacro:macro>
…
```

URDF FILE (generated from the xacro):

```
<?xml version="1.0" ?>
<!--
=================================================================================
===== -->
<!-- |    This document was autogenerated by xacro from
four_dof_arm.xacro             | -->
<!-- |    EDITING THIS FILE BY HAND IS NOT
RECOMMENDED                                  | -->
<!--
=================================================================================
===== -->
<robot name="four_dof_arm">
  <link name="base_link">
    <visual>
      <origin rpy="0 0 0" xyz="0 0 0"/>
      <geometry>
        <box size="1.0 1.0 0.3"/>
      </geometry>
    </visual>
    <collision>
      <origin rpy="0 0 0" xyz="0 0 0"/>
      <geometry>
        <box size="1.0 1.0 0.3"/>
      </geometry>
    </collision>
  </link>
```

The inside “<visual>< visual />” xml tags we will define the geometrical shapes, and inside the “<collision> <collision/>” we will describe the collisional volume the shape should have.

Question 3 – Joint Definition

From the same XARCO/URDF file we can understand that the joint code, describes the parent child relation of the links we have in the robot arm, also we can assign the limits of rotation for each joint.

Question 4 – Assembly Validation

The whole project is a newly created package where the robot model is designed in the URDF/XACRO file, and used a display.launch.py file to easy run the package to visualise the model

Folder Structer:

```
Four_dof_arm/
|-------- launch/
|         |------- Display.launch.py
|-------- urdf/
|         |------- four_dof_arm.urdf
|         |------- four_dof_arm.xacro
|-------- rviz/
          |------- four_dof_arm.rviz
```

The validation of the urdf file is actually checked by the command:

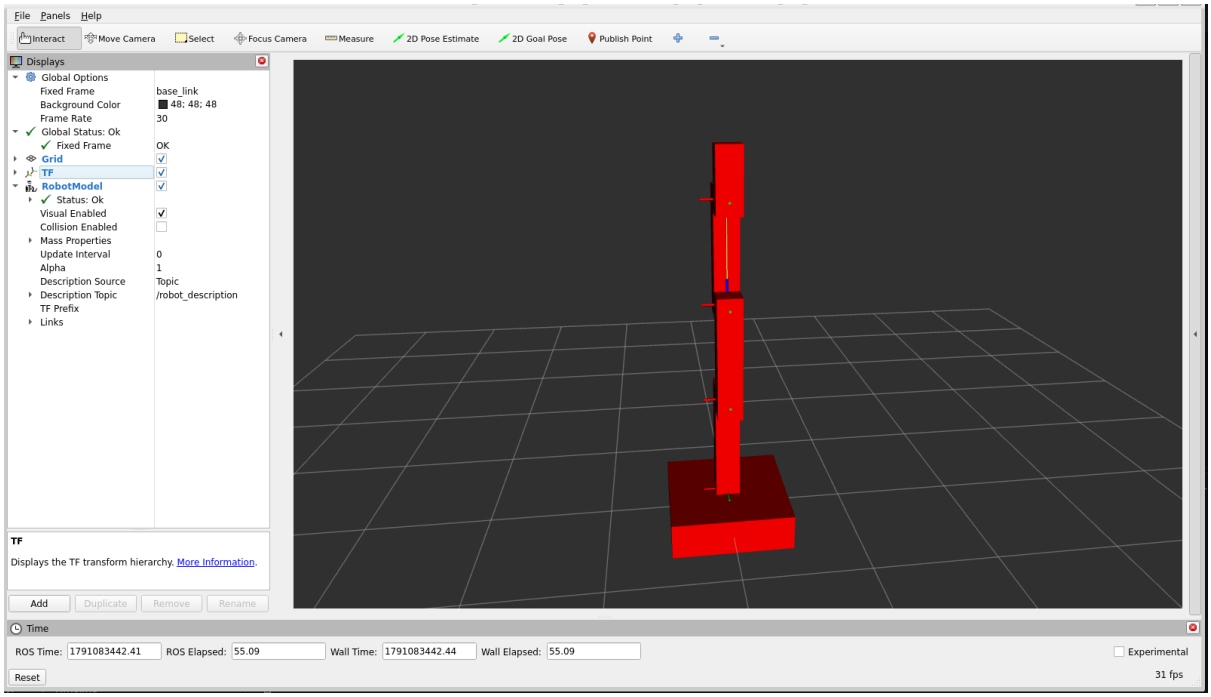
“check_urdf ~/path_to_urdf_file”

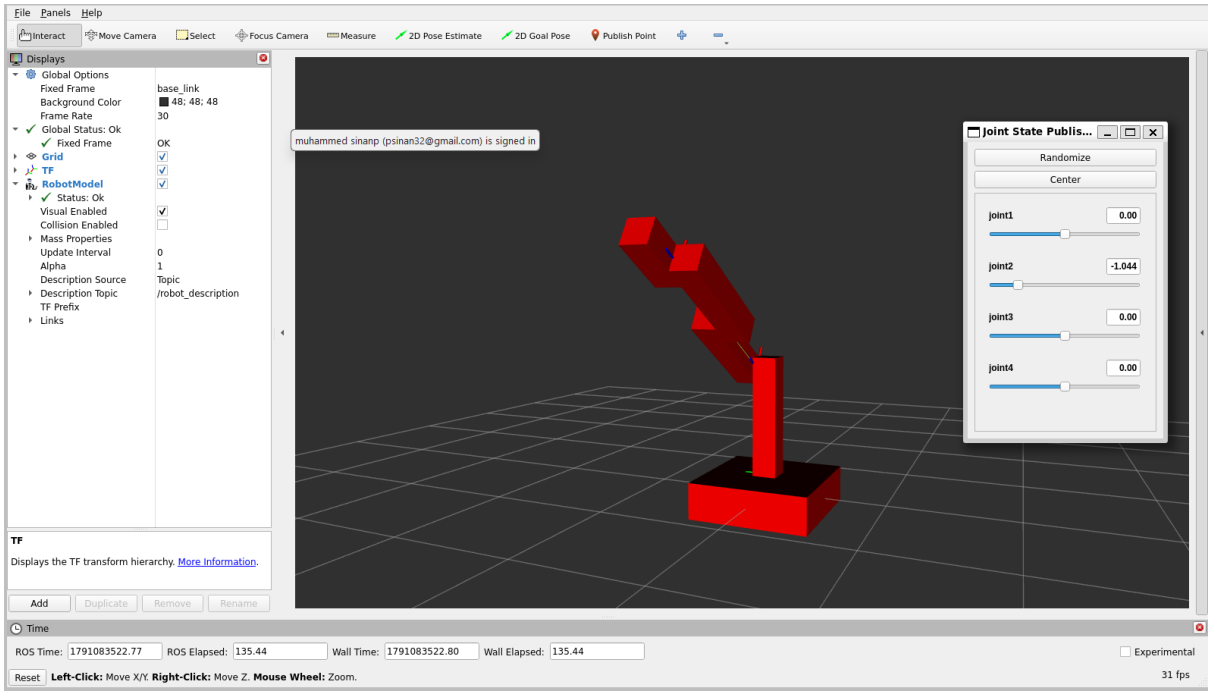
The answer to this will be like:

```
robot name is: four_dof_arm
---------- Successfully Parsed XML ---------------
root Link: base_link has 1 child(ren)
   child(1):  link1
      child(1):  link2
         child(1):  link3
            child(1):  link4
```

Question 5 – RViz Visualization

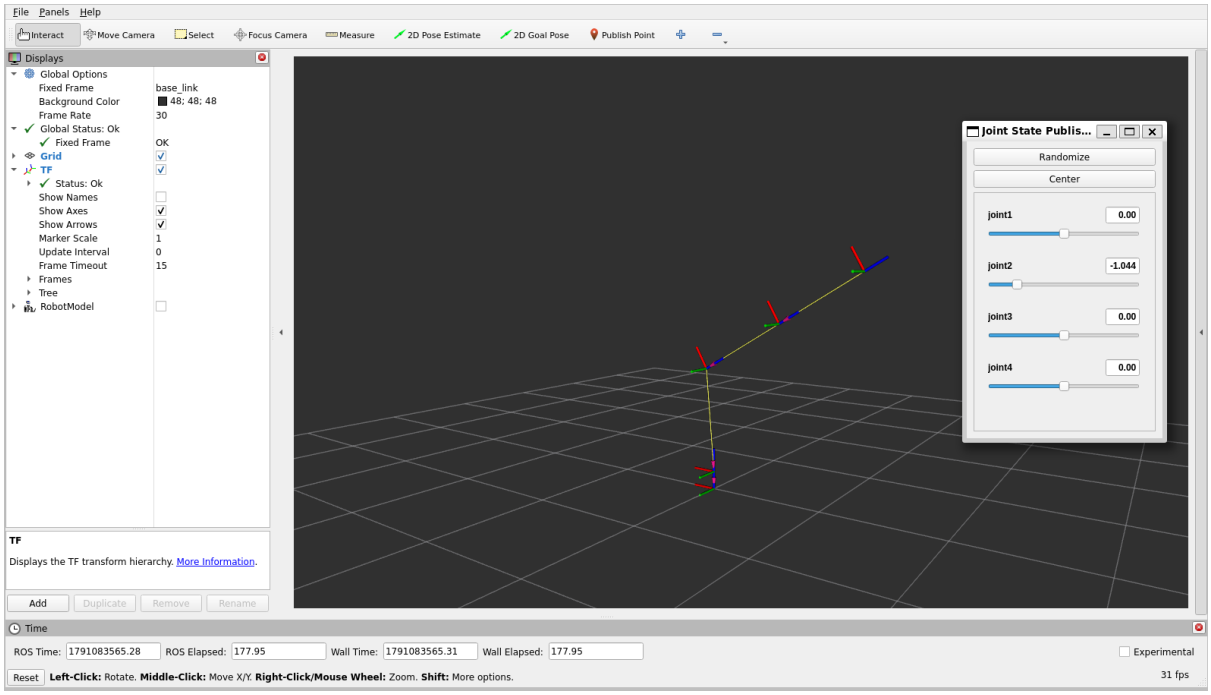
The RViz tool is one of the ros2 tools to visualize the robot models we created in the URDF files, also allows to read states from “robot_state_publisher”.

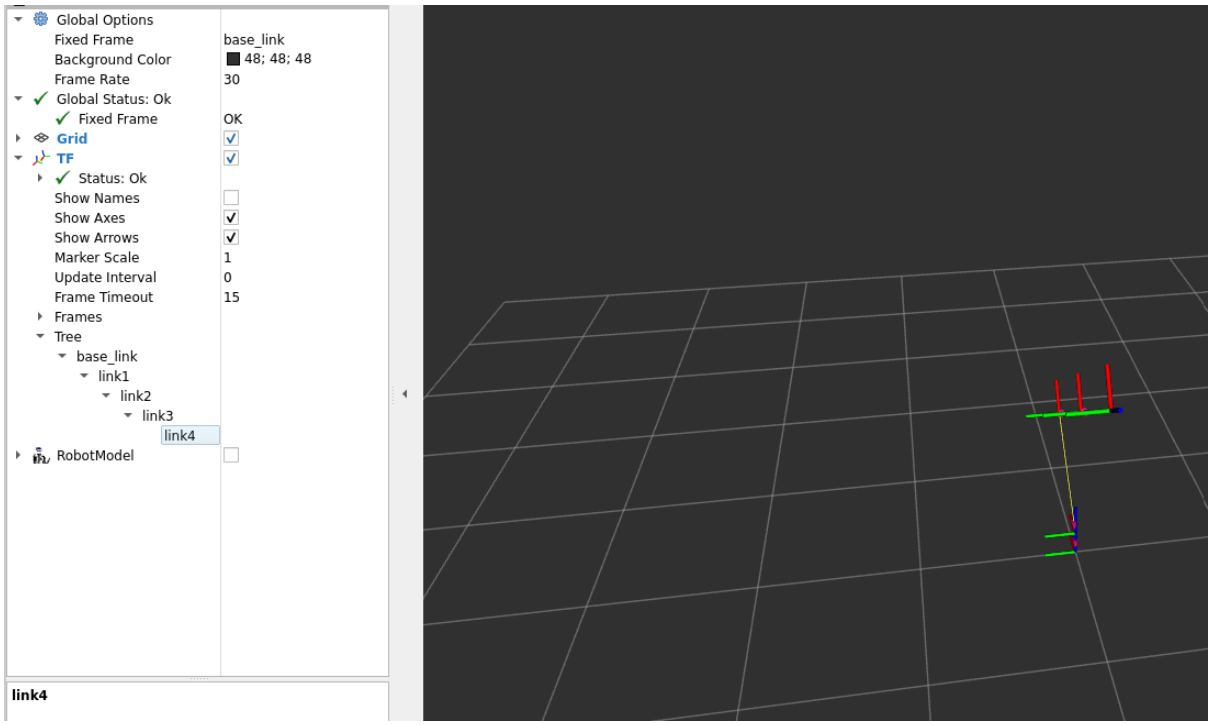




Question 6 – TF Tree Verification

The TF tree is the joints transforms displayed from the data of robot_state_publisher.





Conclusion

The 4-DOF industrial robotic arm was successfully modelled using ROS 2 URDF/Xacro. All links and revolute joints were defined with appropriate geometry, axes, limits, and parent-child relationships. The complete model was validated, visualized in RViz2, and its TF tree was verified using ROS 2 tools.

Appendix

The full codes and packaged it is available in the GitHub repo:
https://github.com/sinanp1x/ros2-4dof-industrial-arm.git